accessories, so it's not very surprising that the first commercial Bluetooth product was an Ericsson hands-free headset. Now, Bluetooth is managed by a large consortium including all the major cell phone manufacturers and many other hardware and software companies, with dozens of products available for a variety of purposes. Bluetooth has a very clearly defined zone of activity—it connects personal devices.

When two Bluetooth devices are brought within range of each other they automatically negotiate a network. More devices can be added to the network or removed at will. One of the devices becomes the Master and regulates the traffic over the ad-hoc network. Up to eight devices can form one network, sharing communications and bandwidth in a configuration called a 'piconet'—a tiny network. In any given area there can be multiple such piconets, and where they need to communicate with other piconets, they can do so across designated devices like gateways. These larger networks are called 'Scatternets'. Bluetooth also operates in the 2.4 GHz frequency band, but it uses the band in a Frequency Hopping mode. It uses 79 1-MHz bands, with each transmission lasting less than a second over any one band. The next transmission is over another band. Thus by rapidly jumping around, Bluetooth avoids interference problems between piconets or other devices in the same band.

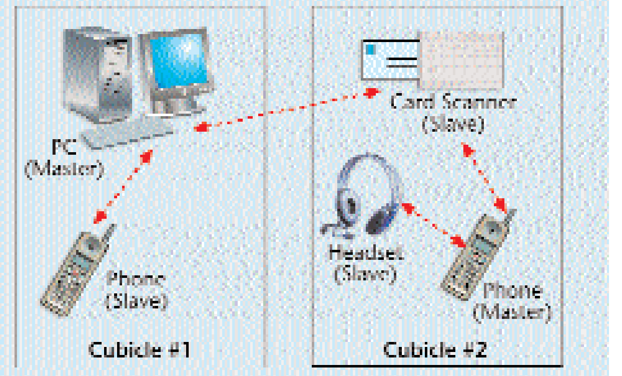
With a normal range of 10 metres and peak transfer speeds below 800 KBps Bluetooth isn't designed for much more than replacing local wires. That doesn't stop people from trying though, and there is a wide misconception that Bluetooth will replace all your wireless networking needs. Obviously, that is not possible. But to prove the point, Toshiba has a portable storage device that connects via Bluetooth. It will be excruciatingly slow, but it's a concept. The 5 GB hard disk based appliance will work out-of-the-box—just place it in the same room as a Bluetooth equipped computer or a network. And Bluetooth-enabling your computer could be as easy as plugging in a key-sized USB device from D-Link. More to the point are Bluetooth-enabled PDAs, laptops and cell phones. Equipped thus, the road warrior could be online via the cell phone still in his pocket—no hassles with IR ports needing careful aiming and alignment.

The problem with Bluetooth is that it demands a chip in every device, which balloons up the cost. As volumes ramp up, costs are expected to come down to more affordable levels in a year or two, but be prepared to pay a premium for the convenience it offers until then.

Home free with HomeRF

HomeRF tries to replace all cables in the home except the power lines. Based on a combination of DECT voice technology, and 802.11 networking, it aims to connect phones, entertainment devices, such as your TV and stereo systems, and your information access systems, computers and Internet gateways. With such a combination, you could route connections from your phone to your PC to your TV set-top box. You could even use the DECT cordless phone to talk to your PC, as a mobile headset replacement. With

A Scatternet



Your phone address book synchronises with your PC over your personal piconet, and can accept contact information your colleague just scanned in from his card scanner via a Scatternet between his piconet and yours

Drive-by Hackings

According to an International Chamber of Commerce report, 92 percent of the 5,000 wireless networks in London freely hand out connections to all comers, providing Internet access and transferring data unencrypted even though there's rudimentary encryption built into all WiFi (Wireless Fidelity) devices. Interestingly, the firm performing the survey gathered data by simply driving around the streets of London with a WiFi laptop and a long-range antenna. Even more interestingly, such long-range antennae can be made out of any suitable metallic tube (e.g. cans of Pringles wafers).

In the early days of computer networking, uninformed companies had banks of modems waiting for any incoming connections. Hackers searching for these open connections would scan entire area codes, a process commonly called 'war-dialling'. The 'in' thing now is 'war-driving'. The equipment is commonplace and affordable, the bounties rich and easy. It is quite common for a network to have firewalls on the Internet connection, yet the backdoor of the wireless network provider has unrestricted access to the backbone infrastructure. People often do not realise the extent that a wireless network can spill over, well outside the buildings it's installed in. Two researchers from Kansas University came up with a technique for mapping wireless network signals (***www.ittc.ku.edu/wlan/***) producing images like the one you see here of a commercial network.